

Technical requirements for PV modules used in grid solar power plants

PV module qualification. PV modules used in grid solar power projects must qualify to the latest edition of any of the following IEC PV module qualification tests or equivalent BIS standards:

1. Crystalline silicon solar cell modules—IEC 61215
2. Thin-film modules—IEC 61646
3. Concentrator PV modules—IEC 62108

In addition, PV modules must qualify to IEC 61730 for safety qualification testing. For PV modules to be used in a highly corrosive atmosphere throughout their lifetime, they must qualify to IEC 61701.

Authorised test centres. PV modules must be tested and approved by one of the IEC authorised test centres. In addition, a PV module qualification test certificate as per IEC standard, issued by Electronics Test & Development Centre (ETDC), Bengaluru or Solar Energy Centre will also be valid. The ministry will review the list of authorised testing laboratories/centres from time to time.

Warranty. Mechanical structures, electrical works and overall workmanship of grid solar power plants must be warranted for a minimum of five years.

PV modules used in grid solar power plants must be warranted for output wattage, which should not be less than 90 per cent at the end of ten years and 80 per cent at the end of 25 years.

Identification and traceability. Each PV module used in any solar power project must use an RF identification tag. The following information must be mentioned in the RFID used on each module (this can be inside or outside the laminate, but must be able to withstand harsh environmental conditions):

1. Name of the PV module manufacturer
2. Name of the solar cells manufacturer
3. Month and year of the manufacture (separately for solar cells and module)
4. Country of origin (separately for solar cells and module)
5. I-V curve for the module
6. Wattage, Im, Vm and FF for the module
7. Unique serial no. and model no. of the module
8. Date and year of obtaining IEC PV module qualification certificate
9. Name of the test lab issuing IEC certificate
10. Other relevant information on traceability of solar cells and module as per ISO 9000

All grid solar PV power plants must install necessary equipment to continuously measure solar radiation, ambient temperature, wind speed and other weather parameters and simultaneously measure DC power generation as well as AC power generated from the plant. They are required to submit this data to NVVN and Ministry of New and Renewable Energy or any other designated agency on line and/or through a report on regular basis every month for the entire duration of the PPA.

Land. The project can be located in a state-designated area (given the advantage of sharing the common infrastructure that is available or will be made available) or a private land, provided the land meets the required criteria for setting up solar power projects.

In addition, developers need to do prospecting using geographical information systems covering parameters like annual average solar radiation levels, protected areas like forests, water bodies, land use, highways and proximity to transmission lines. Based on this prospecting, physical survey can be done for site selection. Thereafter, detailed solar resource assessment studies have to be done through satellite data and computer models to obtain the

hourly solar radiation data to simulate the estimated generation. This process is a must for megawatts-size power projects (even if a state has selected some area for solar farms) as otherwise the power projects' generation cannot be assured.

If the estimated power generation calculations are not accurate, there could be problems for the project approval by the power purchasers (NVVN) besides problems with bankers for sanction of the loan, or even at post-commissioning stage.

About 5 acres of land per MW for solar PV (crystalline) is required. It is assumed at 4-5 acres for crystalline silicon (c-Si) technology and 7-8 acres per MW for thin-film solar (a-Si or CdTe) technology. In reality, it depends on other parameters like the cost of land, ground coverage ratio and choice of sun tracking systems (with sun trackers, the land required is only about 6 acres per MW for crystalline solar modules). To avoid inter-array shading, ground coverage ratio can be 0.45 to 0.65. Power generation will vary based on it. It is preferable to have a plain landscape. If the terrain is uneven, topographical survey is mandatory to take decision as to how the layout is to be done. It is preferable to have a rectangular or square land configuration.

Investment. With the present-day prices, the investment hovers around Rs 100-120 million/MWp depending upon the technology chosen.

Technology. There are several types of semiconductor technologies currently in use for PV solar panels. Crystalline silicon and thin-film are the most widely adopted.

Crystalline silicon. Crystalline silicon panels are constructed by first putting a single slice of silicon through a series of processing steps, creating one solar cell. These cells are then assembled together in multiples to make a solar panel. Crystalline silicon, also called water silicon, is the oldest and the most widely used material in commercial solar panels. There are two main types of crystalline silicon panels:

Mono-crystalline silicon. Mono-crystalline (also called single-crystal) panels use solar cells that are cut from a piece of silicon grown from a single, uniform crystal. Mono-crystalline panels are among the most efficient, yet most expensive on the market. These require the highest-purity silicon and have the most involved manufacturing process.

Multi-crystalline silicon. Multi-crystalline (also called polycrystalline) panels use solar cells that are cut from multifaceted silicon crystals. These are less uniform in appearance than mono-crystalline cells, resembling pieces of shattered glass. They are the most common solar panels on the market, being less expensive than mono-crystalline silicon. They are also less efficient, though the performance gap has begun to close in recent years.

Thin-film. Thin-film solar panels are made by placing thin layers of semiconductor material onto various surfaces, usually on glass.

The term thin-film refers to the amount of semiconductor material used. It is applied in a thin film to a surface structure, such as a sheet of glass. Contrary to popular belief, most thin-film panels are not flexible. Overall, thin-film solar panels offer the lowest manufacturing costs, and are becoming more prevalent in the industry. There are three main types of thin films used: cadmium telluride (CdTe), amorphous silicon, and copper, indium, gallium, selenide (CIGS).

CdTe. CdTe is a semiconductor compound formed from cadmium and tellurium. CdTe solar panels are manufactured on glass. These are the most common type of thin-film solar panel on the market